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Laser-ablative engineering of phase singularities in plasmonic metamaterial arrays for biosensing applications

Andrey I. Aristov,¹ Urs Zywiets,² Andrey B. Evlyukhin,² Carsten Reinhardt,² Boris N. Chichkov,² and Andrei V. Kabashin^{1,a)}

¹Aix Marseille University, CNRS, LP3 UMR 7341, Campus de Luminy - Case 917, 13288, Marseille Cedex 9, France

²Laser Zentrum Hannover e.V., Hollerithallee 8, 30419 Hannover, Germany

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By using methods of laser-induced transfer combined with nanoparticle lithography, we design and fabricate large-area gold nanoparticle-based metamaterial arrays exhibiting extreme Heaviside-like phase jumps in reflected light due to a strong diffractive coupling of localized plasmons. When employed in sensing schemes, these phase singularities provide the sensitivity of 5×10^4 deg. of phase shift per refractive index unit change that is comparable with best values reported for plasmonic biosensors. The implementation of sensor platforms on the basis of such metamaterial arrays promises a drastic improvement of sensitivity and cost efficiency of plasmonic biosensing devices. © 2014 AIP Publishing LLC. [<http://dx.doi.org/10.1063/1.4865553>]

Plasmonic biosensors have demonstrated spectacular progress during the last years and became an important tool for the detection and studies of biological binding events between a target analyte (antigen, DNA, protein, etc.) and its corresponding receptor (antibody, DNA capture, ligand, etc.).^{1,2} Conventional Surface Plasmon Resonance (SPR) biosensors are based on the excitation of Surface Plasmon Polaritons (SPPs) over a 50-nm gold film in the Kretschmann-Raether prism geometry to control biomolecular interactions by refractive index (RI) monitoring.^{3–5} Here, amplitude characteristics, such as the resonant angle of light incidence^{3,4} (or resonant wavelength⁵), are typically considered as the signal parameter. An extension of SPR, called Localized Plasmon Resonance (LPR), uses plasmonic gold-based nanostructures, which can add new functionalities, including the compatibility with modern bio-molecular nano-architectures,^{6,7} size selectivity and spectral tuneability,^{6,8,9} strong field enhancement,¹⁰ etc. However, both SPR sensing configurations are still lacking of sensitivity to compete with classical labelling methods. The detection limit in terms of the amount of a biomaterial accumulated on the surface is typically lower than 1 pg/mm^2 and 1000 pg/mm^2 for SPR and LPR, respectively, that is inferior to labelling methods by about 2 orders of magnitude.

We have demonstrated a way for a radical improvement of sensitivity of plasmonic biosensing technology by employing phase properties of light instead of the amplitude ones.^{11–13} The use of phase as the sensing parameter in conventional SPR geometry makes possible the decrease of the detection limit by up to two orders of magnitude.¹⁴ Further improvement of phase-sensitive sensors becomes possible using diffractive coupling of localized plasmons^{15–18} in photonic particle arrays of metamaterials. It was recently shown¹⁹ that a proper engineering of these resonances can provide a drastic drop of intensity in the resonant point (light “darkness”), which is accompanied by an extreme phase

jump in the point of darkness. The employment of this jump in sensing schemes could lower the detection limit down to single molecular level, but the implementation of such phase singularities imposes tough requirements on the quality of particle array metamaterials. Such high quality can generally be achieved through the involvement of expensive and time-consuming methods, such as E-beam lithography (EBL) and Focusing Ion Beam (FIB), which complicates application prospects of photonic metamaterial components. In addition, it is difficult to produce large arrays with these methods, which is required for the development of low-noisy multi-channel phase-sensitive instrumental schemes.¹⁴

Here, we examine phase features using large area plasmonic nanoparticle arrays fabricated by cost-efficient laser-assisted methods based on Laser Induced Transfer (LIT)^{20–22} from nanosphere lithography-prepared flat triangular-prism-type metallic structures. These laser-fabricated arrays are known to exhibit collective plasmonic resonances in transmission spectra under normal incidence of pumping light beam.²² By monitoring reflection spectra under an oblique incidence of the light beam on metamaterial structure, we reveal sharp phase features in reflected light, which can be employed in phase-sensitive plasmonic biosensing.

The fabrication procedure consists of two steps, as illustrated in Figure 1. First, a hexagonal structure of gold triangles (triangular prisms) is fabricated on a glass (donor) substrate using nanosphere lithography.⁶ This method is based on physical vapour deposition of a thin 45 nm gold film over a lithographic mask consisting of an array of mono-dispersed colloidal spheres. After lift-off of the colloidal mask, a hexagonal array of plane triangular-shaped nanoparticles remains on the substrate surface (see Figure 1(b)). The second step implies laser-induced transfer of the triangular structures from the donor onto a receiver substrate. Femtosecond laser pulses irradiate the back side of the donor substrate, as shown schematically in Figure 1(a). The nanoparticle structures are fabricated at the laser fluence of 0.06 J/cm^2 with 1 kHz repetition rate and 1 mm/s sample translation speed. The triangular structures are melted by

^{a)} Author to whom correspondence should be addressed. Electronic mail: kabashin@lp3.univ-mrs.fr. Telephone: +33 (0) 4 91 82 93 83.

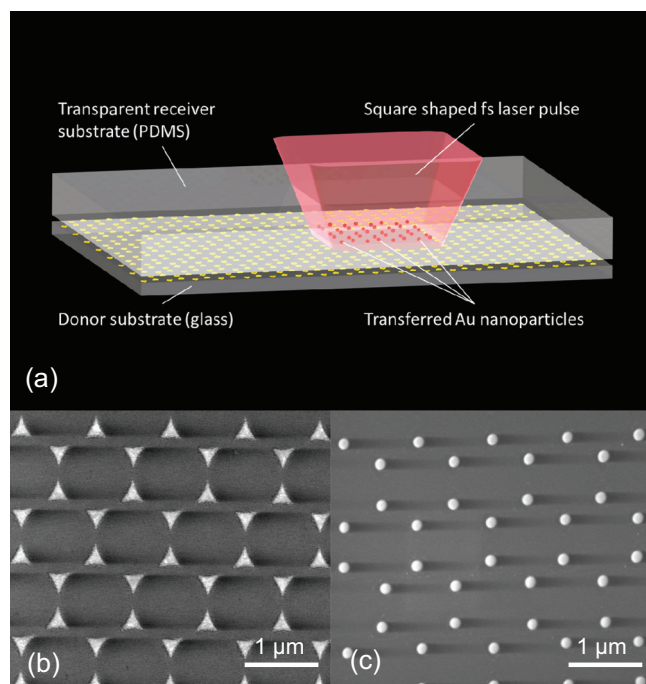


FIG. 1. (a) Schematic representation of laser induced transfer procedure for the fabrication of Au nanoparticle array; (b) SEM images of nanoparticle structures before (b) and after (c) laser induced transfer. Due to the surface tension, the shape of the nanoparticles is changing from flat triangular-prism-type to spherical.

laser irradiation and transformed into spherical droplets by strong surface tension forces. Droplets are ejected from the donor towards the elastic polymer polydimethylsiloxane (PDMS) receiver substrate due to inertia of the center-of-mass movement, which is initiated by the transformation of the flat triangular prisms into spheres with a diameter of approximately 110 nm. Figure 1(c) shows a fragment of a typical gold nanoparticle array produced by the described laser-assisted procedure. The unit cell size, i.e., the length of the longest diagonal in the hexagonal unit cell, is approximately 1130 nm. The laser-assisted procedure enables fast and easy fabrication of relatively large arrays (up to 1 mm²) with a relatively uniform distribution of hexagonal gold nanoparticles on the PDMS substrate. The size of the “ideal” defect-free zone on the sample is estimated as 30 × 30 μm², which exceeds the minimal metamaterial size for an efficient excitation of diffractive-coupled plasmon modes.¹⁹

After the laser-assisted material transfer and the formation of an array of spherical features, the metamaterial exhibited narrow resonances under normal incidence of white light beam onto the nanoparticle array similarly to how it was observed in Ref. 22. As shown in Figure 2(a), the spherical nanoparticle array shows Fano-type resonances²³ at 694–696 nm and 712–714 nm, when the structure is in contact with air ($n_{\text{air}} = 1$) and water ($n_{\text{wat}} = 1.33$) environments, respectively (notice that a slight difference in dip intensities for orthogonal polarizations is explained by a misalignment of the optical measurement setup related to main metamaterial axes). While the resonance width is relatively broad for air environment (>70 nm full width at half maximum (FWHM)), the width of the resonance becomes much narrower (~30 nm) for water ambience. The narrowing effect is

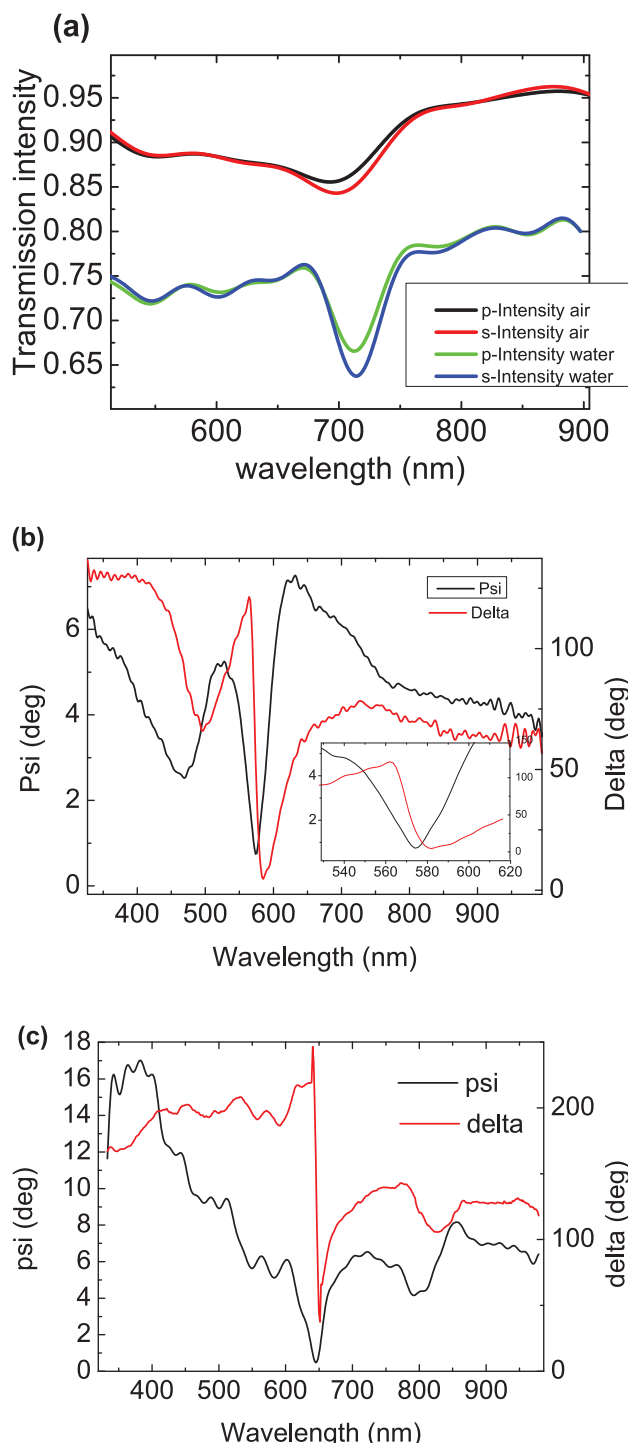


FIG. 2. (a) Transmission spectra at normal incidence for orthogonal polarizations when the nanoparticle array is in contact with air (black and red curves) and water (blue and green curves) environment. Spectral dependences of ellipsometric parameters Ψ (black curves) and Δ (red curves), while the nanoparticle structure is in contact with air with 57° of light beam incidence (b) and water (c) environments. The experiment with water environment was done in Attenuated Total Reflection geometry with 40° incidence.

consistent with the diffractive coupling of the localized plasmon resonances of the individual particles when one of the diffraction orders starts to propagate along the nanoparticle array surface.^{15–19,24–26} It should be noted that in most cases the excitation of diffractive-coupled LPR requires the embedding of nanoparticles in a uniform (symmetric)

environment where refractive index is identical above and below the array plane. The ultra-narrow resonances often disappear in an asymmetric environment.²⁶ Although in our case the effect of narrowing was absent for the sample in air environment corresponding to an essentially asymmetric case ($n_{\text{pdms}} = 1.47 > n_{\text{air}} = 1$), the narrow resonances still appeared when the structures were in contact with water ($n_{\text{air}} = 1.33$).

It is important that under normal incidence there is no significant difference in optical responses for orthogonally polarized components (Figure 2(a)), since only the component of the electric field vector over the array plane is present. Therefore, any possible phase jump in the beam appears to be nearly identical for the orthogonal components and can be measured only by relatively complicated interferometric methods implying the use of a reference beam, which is not passing through the structure, and a highly coherent light source (similarly to how it was done in Ref. 11). However, when the pumping light beam is directed at some angle to the array plane, a p-polarized electric field component perpendicular to the array plane appears. It is known^{15,17,19,27} that ultra-narrow collective resonances can also be efficiently excited by p-polarized light under this oblique geometry and observed at zero reflection order. As the optical response of the structure to s-polarized light is essentially different under these conditions with no resonant feature, this component can be used as the reference one to monitor the phase properties. In this case, since both polarization components have the same optical path, phase can be tracked by relatively simple ellipsometry or common path interferometry tools using non-coherent light sources. In our experiments, we used a Woollam M-2000 spectroscopic ellipsometry system. Ellipsometry provides information on the amplitude of reflected light determined by the equation $r_p/r_s = \tan\Psi e^{i\Delta}$, where r_p and r_s are reflectivities of p- and s- components, respectively. Additionally, the difference of phases Δ of p- and s- components can be determined by means of a rotating analyser. A white light beam was reflected from the nanoparticle arrays at different angles (20° – 80°), while the ellipsometric parameters were recorded in the spectral range of 350–700 nm. Figures 2(b) and 2(c) show typical spectral dependencies of the ellipsometric parameters Ψ and Δ for an angle of incidence of 57° , when the nanoparticle structures were in contact with air and water environments, respectively. One can see that despite essentially asymmetric environments (the refractive index of the substrate is $n = 1.47$ and did not match that of air and water), we were able to observe narrow resonances with the resonance width of less than 20 nm FWHM. In this case, the main resonances were blue-shifted related to the normal incidence case (Figure 2(a)) and observed around 470 nm and 570 nm for air and water environments, respectively. It should be noted that the efficient generation of the narrow resonances even under a complete asymmetry of the environment is a characteristic feature of diffraction-coupled LPR excited by p-polarized light.^{15,17,19} It is interesting that in addition to the main resonances at 574 and 645 nm, associated with the diffraction coupling at the effective lattice constant of $L \approx 849$ nm, we could observe additional coupling LPR modes. Although these modes were not clearly distinguishable when the

structure was in contact with air environment, we could observe additional small peaks at 550 nm, 790 nm, and 990 nm in the water case (see Figure 2(c)). It is clear that the appearance of these additional modes is related to a relatively complex geometry of hexagonal arrays, which can simultaneously provide several lattice constants and thus support different diffractive orders associated with these constants. However, the most striking observation consists in the presence of a sharp phase jump at the point of LPR at 645 nm (Figures 2(b) and 2(c)). The total phase variation was about 120° and 180° in the case of air and water environments, respectively. As shown in the high resolution plot of the inset in Figure 2(b), the phase singularity occurs in the minimum of the resonance curve, which is consistent with the previous data.^{11,13} As we earlier explained,^{19,28} the sharpness of the phase jump is related to the “darkness” of the resonance rather than to its width. Here, the lower the intensity in the resonance minimum is, the sharper is the phase feature. As shown in Figures 2(b) and 2(c), laser-fabricated structures provide resonances with very low light intensity in the minimum (less than 1%), which enabled relatively sharp phase jumps.

To assess the efficiency of the laser-fabricated structures in phase-sensitive sensing, we carried out a series of model sensing tests using a well-established glycerine model. In this model, different concentrations of glycerine are added to water in order to provide controllable changes of the refractive index of the aqueous environment. Aqueous solutions of glycerine were pumped through a flow cell by a peristaltic pump and brought into contact with the nanoparticle array, while it was illuminated in Attenuated Total Reflection geometry using a glass prism (the prism was in immersion contact with the sample slide), as shown in Figure 3(a). The block containing the prism, the metamaterial slide with nanoparticle array, and the flow cell was put onto the ellipsometer platform and examined.

The inset of Figure 3(b) shows the response of amplitude parameter Ψ under the addition of 2.5%, 5%, 7.5%, and 10% of glycerine into the solution. It is visible that an increase of RI due to the glycerine addition leads to a red-shift of the resonance. Figure 3(b) shows the LPR position as a function of the refractive index of the medium, as conditioned by different concentrations of glycerine. As shown in the figure, the resonant position linearly shifts under a relatively wide range of refractive index variations (~ 0.01 RIU). The slope of the linear dependence evidences the sensitivity of 290 nm/RIU, which looks comparable with the best values of sensitivity reported for localized plasmons,⁸ but still it is much weaker than spectral responses of thin film-based SPR (1000 nm/RIU).^{1,2,5} An important comparative parameter of sensing devices is the figure of merit $FOM = (\Delta\lambda/\Delta n)(1/\Delta\omega)$ (where $\Delta\omega$ is the full-width of the resonant dip at half-maximum and $\Delta\lambda$ is the resonance shift for Δn refractive-index change), which takes into account the sharpness of the resonance and thus examines the ability to sensitively measure tiny wavelength changes.⁸ In our case, $FOM = 15$, which is much better compared to uncoupled LSP-based sensors ($FOM = 8$)⁸ due to much narrower resonance width of diffraction-coupled SPR. However, the most interesting result addresses prominent

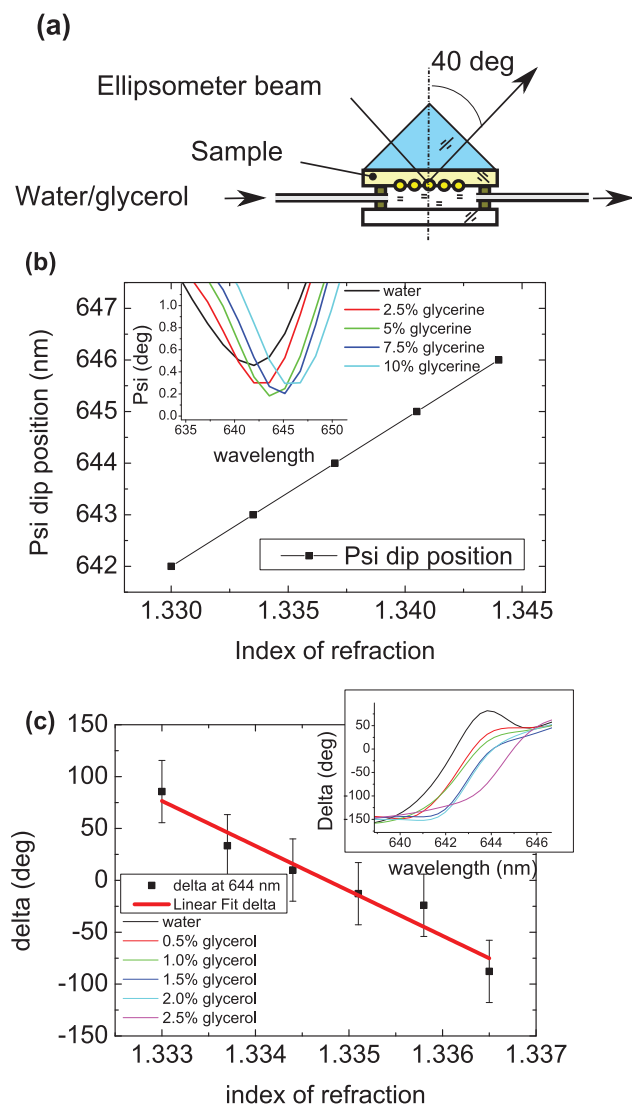


FIG. 3. (a) Schematics of experiments simulating RI changes; (b) amplitude parameter Ψ as a function of RI of water/glycerol solutions contacting the metamaterial array. Inset: Typical spectral dependences for Ψ under different RI corresponding to different concentrations of glycerol in water; (c) magnitude of phase Δ at 644 nm as a function of RI. Inset: Typical spectral dependences for Δ .

phase responses. Figure 3(c) shows the value of phase Δ as a function of RI at a fixed wavelength of 664 nm under a relatively small range of RI variations (0.0025 RIU) corresponding to dynamic range of phase variation. It is clear that phase demonstrates a similar linear dependence to RI changes within this range. Here, even a very small refractive index change of 10^{-3} RIU leads to a drastic phase shift of 5° , yielding to the sensitivity of about 5×10^4 deg./RIU. It should be noted that this value of phase sensitivity is close to the best phase sensitivities reported in conventional SPR (10^5 deg./RIU).^{11,13} Thus, despite a considerable handicap in terms of amplitude sensitivity compared to conventional SPR, the employment of phase sensitivity completely compensates this gap. Measuring the phase sensitivity at the level of 5×10^4 deg./RIU, we can now estimate the low detection limit (LOD), which characterizes minimal detectable variation of RI contacting the optical transducer. The LOD strongly depends on the level of instrumental and other noises in the system and these noises are known to be

relatively high in our ellipsometric measurements.¹⁴ As was demonstrated in Ref. 13, the resolution of phase measurements for a thermally stabilized system with advanced phase detection can be better than 5×10^{-3} deg. Under this phase resolution, the LOD is estimated to be lower than 10^{-7} RIU, which is comparable or better than in commercial amplitude-sensitive SPR units.¹⁻³ Thus, despite a relatively simple fabrication procedure, the prepared structures exhibited “dark” ultra-narrow resonances, yielding to prominent phase singularities.

To conclude, we used cost-efficient methods of laser-induced transfer combined with nanosphere lithography to fabricate large-area plasmonic metamaterials based on hexagonal arrays of gold spherical nanoparticles. We showed that these nanoparticle metamaterials can provide resonances with nearly zero intensity in reflected light for p-polarized light, yielding to the generation of phase singularities. Being employed in sensing geometries, such singularities can provide an impressive sensitivity of 5×10^4 deg./RIU and the LOD lower than 10^{-7} RIU, which is comparable or better than in LPR and SPR sensors.

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